



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RAJA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

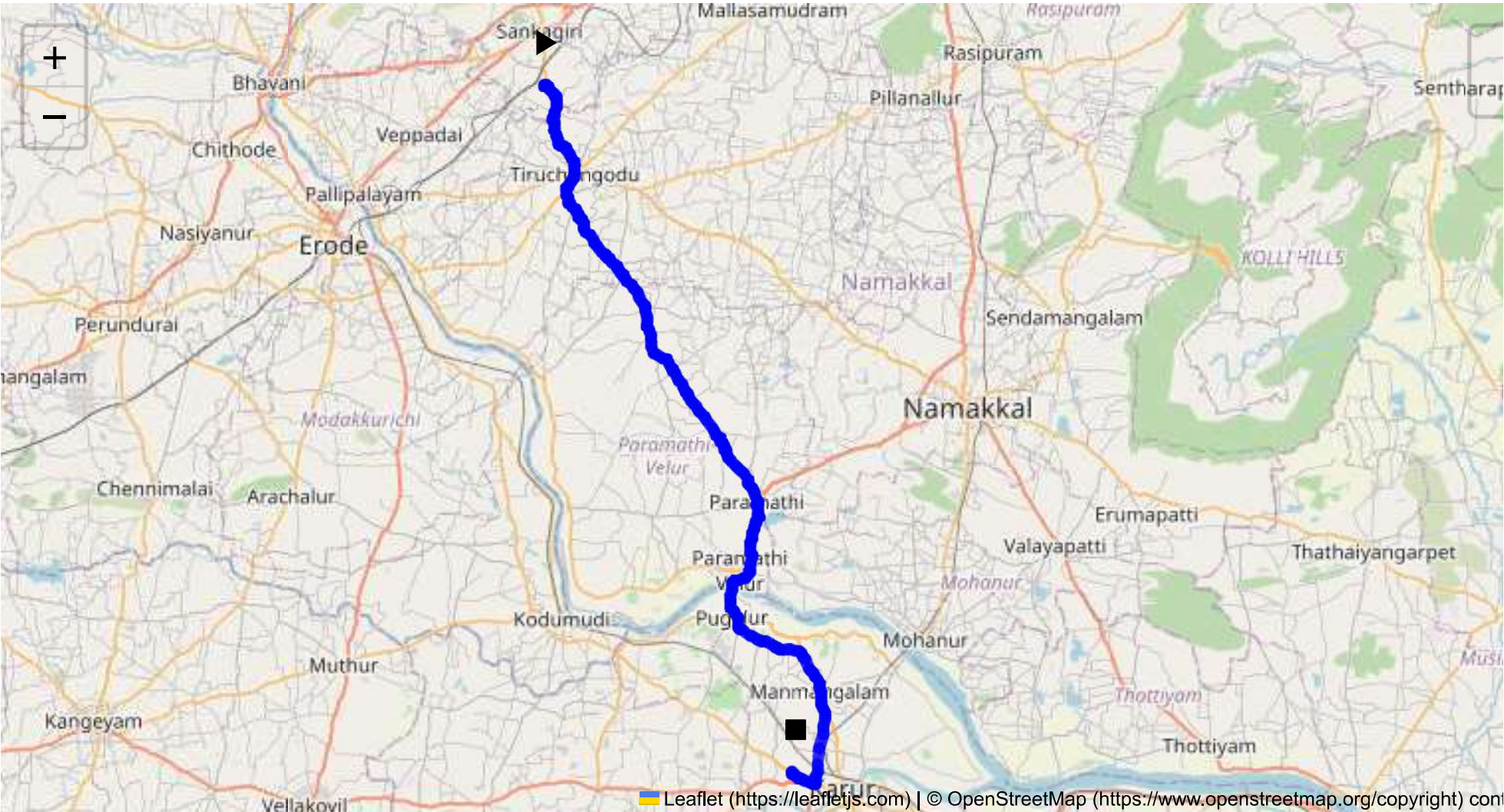
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 66.64 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (10.969188, 78.047228)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

Route Overview:

- The route from CVQF+23W, Sangagiri to X29W+MQX, Reddipalayam spans approximately 66.64 kilometers and typically takes about 1.35 hours for heavy vehicles.

- The journey primarily utilizes National Highways, with stretches of State Highways and local roads.

1. Overview of the Route Map:

- The route spans through rural and semi-urban areas, connecting key points that facilitate the transport of goods between districts.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- **Weather Conditions:** Predominantly tropical, with high temperatures and monsoon rains.
- **Hazards:** Monsoon season (June to September) can present challenges with heavy rainfall leading to waterlogging. Sun glare during peak daylight hours can also affect visibility.

3. Traffic Patterns:

- **Peak Hours:** Typically, morning (8-10 AM) and evening (5-7 PM), when local commuters are traveling to work and returning.
- **Congestion-Prone Areas:** Near towns such as Namakkal and Karur, where local traffic can slow down freight movement.

4. Assessment of Road Quality and Infrastructure:

- **Road Quality:** National Highways are generally well-maintained, but State Highways and rural roads may have varying conditions, with potential potholes and uneven surfaces.
- **Infrastructure:** Adequate signage and lighting on major highways, but local roads may lack sufficient lighting.

5. Suggestions for Alternative Routes for Emergencies:

- In case of roadblocks or accidents, consider the following alternatives:
 - Using NH44 or NH81, which run parallel for certain stretches and can provide detours if necessary.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Permits are required for transporting hazardous materials.
- Restrictions on vehicular movement in core town areas during peak hours for safety reasons.

7. Overview of Historical Incidents:

- Some minor incidents have been recorded due to overloaded vehicles or inadequate securing of loads.
- Occasional accidents caused during the monsoon due to slippery roads.

8. Environmental Considerations and Sensitive Areas:

- Route travels close to agricultural areas; caution needed to prevent spillage and contamination.
- Avoid activities that could disturb local wildlife or protected areas.

9. Communication Coverage:

- Generally good coverage along National Highways.
- Potential dead zones in rural stretches, particularly when venturing off the main highways.

10. Estimated Emergency Response Times:

- Urban areas: 20-30 minutes response time.
- Rural areas: 45-60 minutes or more, depending on proximity to the nearest town with emergency services.

12. Overall Risk Assessment:

- **Risk Level:** Moderate
- **Key Risks:** Include weather-related disruptions, road quality issues on minor roads, and traffic in urban areas.
- **Recommendations:** Operators should be prepared for rapid weather changes and ensure compliance with hazardous transport regulations.

Conclusion:

- The transport route from Sangagiri to Reddipalayam requires careful planning, particularly concerning weather conditions and peak traffic hours.
- By maintaining regular communication with local authorities and emergency services, along with adhering to regulations, risks associated with heavy vehicle transport can be effectively managed.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.96304, 78.06131	15 KM/Hr
2	U-Turn	High	10.9703584, 78.04508489999999	10 KM/Hr
3	Turn	Medium	11.43822, 77.87348	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	High	11.37868, 77.89498	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.15573, 78.02176	15 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr
12	Turn	High	10.96269, 78.06101	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	High	10.96373, 78.05741	15 KM/Hr
14	Turn	High	10.96384, 78.05740	15 KM/Hr
15	Turn	Medium	10.97031, 78.04515	30 KM/Hr
16	Blind Spot	Blind Spot	10.97043, 78.04511	10 KM/Hr
17	Blind Spot	Blind Spot	10.96901, 78.04706	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium
21	clinic	Sivabakkiyan Poly Clinic	10.9642254, 78.0549579	30 km/h	Medium
22	hospital	Krishna Medical Centre Hospital	10.9643009, 78.0541215	30 km/h	Medium
23	hospital	Meganathan Hospital	10.9658739, 78.052368	30 km/h	Medium
24	hospital	vasu medical	10.9638208, 78.0530404	30 km/h	Medium
25	clinic	Pranic Healing Centre	10.9670704, 78.050346	30 km/h	Medium
26	hospital	Sathyam Hospital	10.968865, 78.048055	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96304, 78.06131



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 10.9703584, 78.04508489999999



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96269, 78.06101



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96373, 78.05741



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96384, 78.05740



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.97031, 78.04515



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.97043, 78.04511



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.96901, 78.04706